AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

Faculty of Computer Studies

TM112
Introduction to Computing and Information Technology II

Mid-term Examination (MTA)
Spring 2024/2025

Date: 10 /April /2025

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Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 hours

tructions:

- Total Marks: **30**
- This exam consists of **3 parts**.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: Multiple Choice Questions (5 marks).

1) Using 3 bits, we can code ------- colors
   a. 3
   b. 4
   c. 7
   d. 8
2) Psychoacoustics is the study of how humans perceive and respond to
   a. sound
   b. Image
   c. Video
   d. All of the above
3) Ubiquitous means:
   a. Unavailable around us
   b. Everywhere around us
   c. High cost
   d. Low cost
4) An entity can be:
   a. Tangible
   b. Intangible
   c. A concept
   d. All the above
5) ---------- refers to the ability of a UI to prevent errors if possible, or to make them easy to recover from, if not
   a. Affordance
   b. Simplicity
   c. Tolerance
   d. Feedback

Part 2: Short questions (10 marks)

Question 1: (3 marks)

Give the definition of the following:

a) Hoax message
b) Spam
c) Phishing

Question 2: (4 marks)

For the given numbers find their equivalent in the specified base. **Show your steps.**

$(110101110)_2 = ()_{16}$
$(1A3)_{16} = ()_{10}$
$(3F2)_{16} = ()_8$
$(101110)_2 = ()_8$

Question 3: (3 Marks)

List 3 important tips to remember when participating in a forum?

Part 3: Problem Solving / Long Questions (15 marks)

Question 1: (6 marks)

Consider an image of size 200 x 140. How many bits and bytes are required to store this image inside the computer

a. Using only 2 colors: black and white.

b. Using 3 bits per pixel in grayscale model.

c. Using RGB model.

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Question 2: (4 marks)

a) What is data?

b) List and define 3 common data types used in databases?

Question 3: (5 marks)

List two techniques used in lossless compression and describe each one of them

End of question

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